X-ray detection

Two ways to convert X-rays into charge

Method	Signal path
Indirect conversion	X-rays deposit energy in a scintillator. It emits light that a photodiode converts to charge.
Direct conversion	X-rays directly produce charge in a photodiode. An electric field moves the charge to the pixel electrodes.

A **scintillator** is a material that emits light after absorbing energy.

What does a phosphor do?

A scintillating **phosphor** converts some of the energy deposited by an X-ray into visible light.

- **X-ray stopping.** A thicker layer intercepts more X-rays. Light can also spread farther before reaching the sensor.
- **Light yield.** CsI(Tl), or thallium-doped cesium iodide, produces roughly 54 visible photons per keV deposited.
- **Light collection.** Only some of that light reaches the sensor and generates charge.

A 50 keV X-ray that deposits all its energy in CsI(Tl) produces about **2,700 visible photons**.

If each visible photon carries about 2.25 eV, what fraction of the deposited energy becomes light?

How long does the phosphor keep glowing?

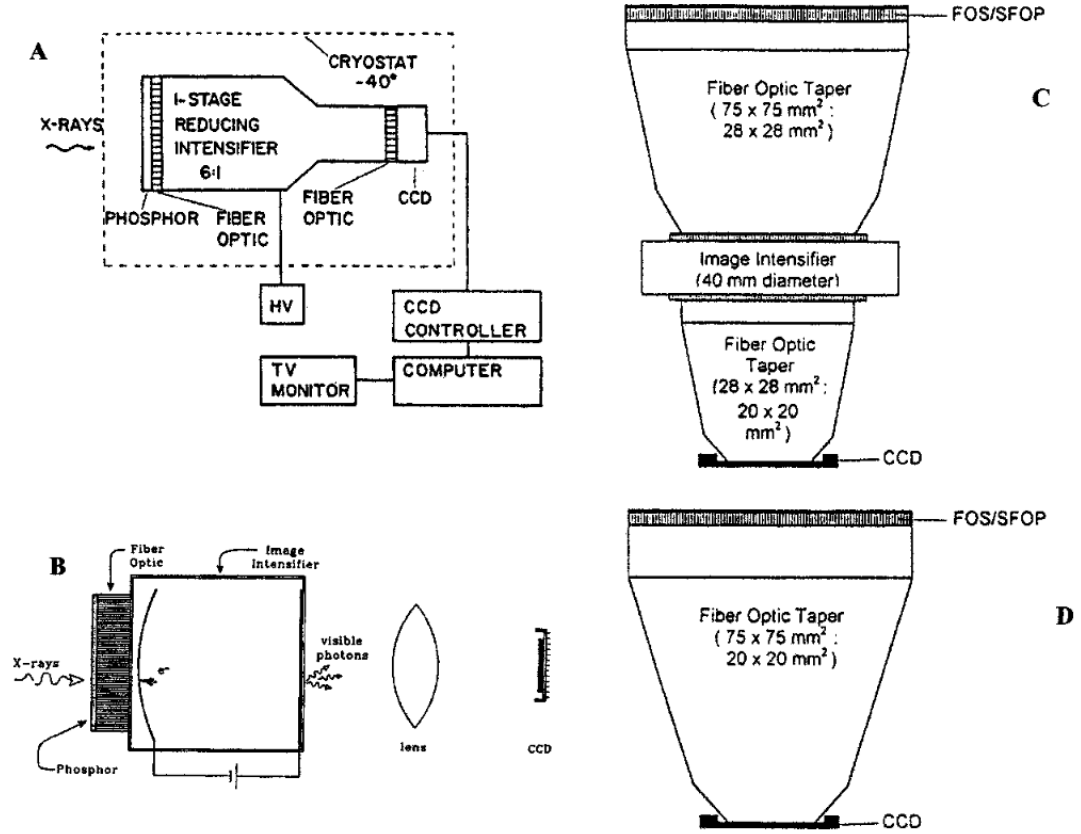
- **Initial decay.** The light output drops rapidly after the interaction. A fast decay component in CsI(Tl) is of order $1\ \mu$s.
- **Persistence, or afterglow.** A weak intensity tail can remain for milliseconds or longer. Trapped energy can be released later as light.

If the first image is bright and the next is dark, what might an x-ray detector built with a persistent phosphor show for the second frame?

How to transfer light to a CCD?

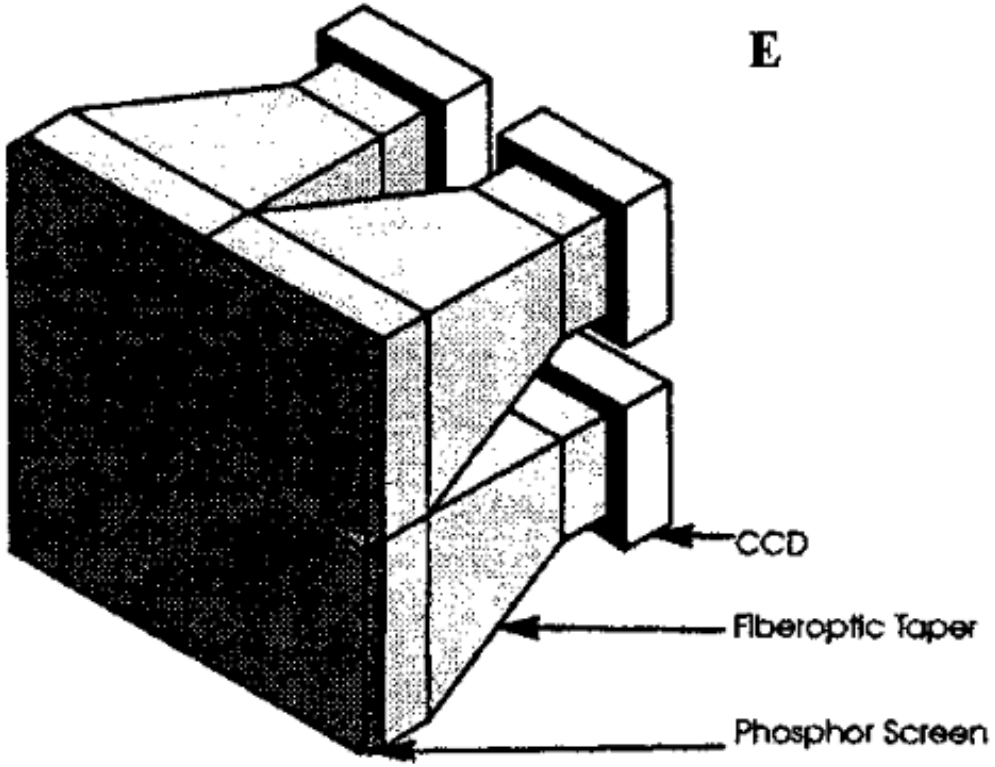
Coupling	What it does
Lens	Focuses an image of the phosphor onto the CCD. The lens collects only part of the emitted light.
Fiber-optic plate	A bundle of fibers guides light from the screen to corresponding positions on the CCD.
Fiber-optic taper	The bundle has a larger entrance than exit. It maps a larger screen area onto a smaller CCD.

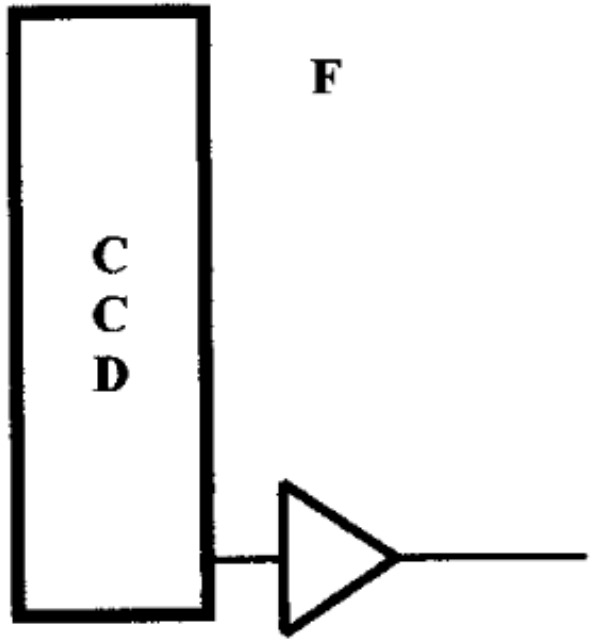
CCD light coupling



Tiled CCD

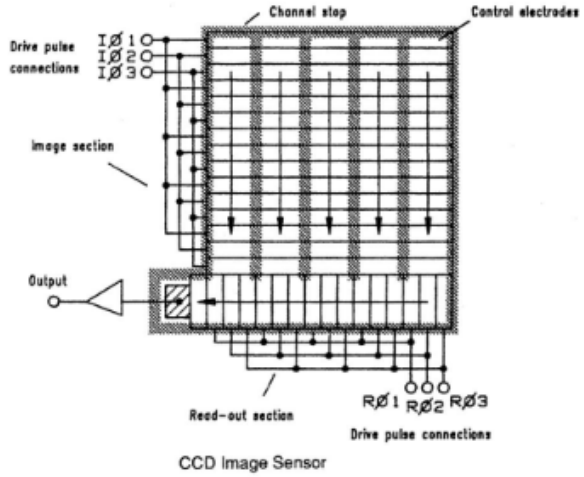
A tiled detector uses several taper–CCD modules side by side. The modules cover a larger area, with seams and geometric differences to correct.

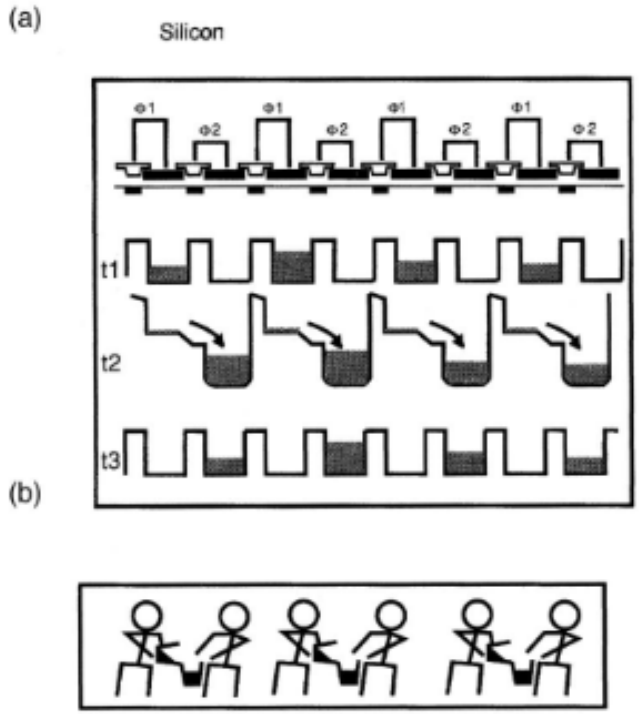




CCDs

A charge-coupled device (CCD) stores the photo-generated charge in pixel wells.





Integration and counting are readout choices

- **Energy integration:** accumulate the detector signal over an exposure. A larger deposited energy generates more charge and a greater output signal.
- **Photon counting:** resolve individual events and register counts. Some systems also sort events into energy ranges.

Direct and indirect conversion describe how charge is made. Integration and counting describe what the readout reports.

How many X-rays interact in a thin silicon wafer?

Take a 300 μm active Si layer, with X-rays entering perpendicular to the wafer. The fraction that interacts is $1 - e^{-\mu t}$.

Photon energy	Si attenuation length $1/\mu$	Interacts in 300 μm
20 keV	0.96 mm	27%
30 keV	3.0 mm	9.5%
60 keV	13.4 mm	2.2%

Discuss: If most photons pass through, would smaller pixels fix the problem?

Amorphous selenium wide-area panel

Amorphous selenium (a-Se) can be deposited as a continuous layer over a large pixel array.

1. An X-ray interaction creates electrons and holes in the selenium.
2. A voltage across the layer drives the charges toward the electrodes.
3. Each pixel stores charge. Thin-film transistors switch the pixels for readout.

A direct mammography panel can cover about 24×29 cm with 70 μm pixel pitch. Thin film transistor electronics allows for these large panels.

Medipix3 counts individual X-rays

Medipix3 is a CMOS readout chip bonded pixel by pixel to a separate semiconductor sensor. Direct detection with digital pulse counting.

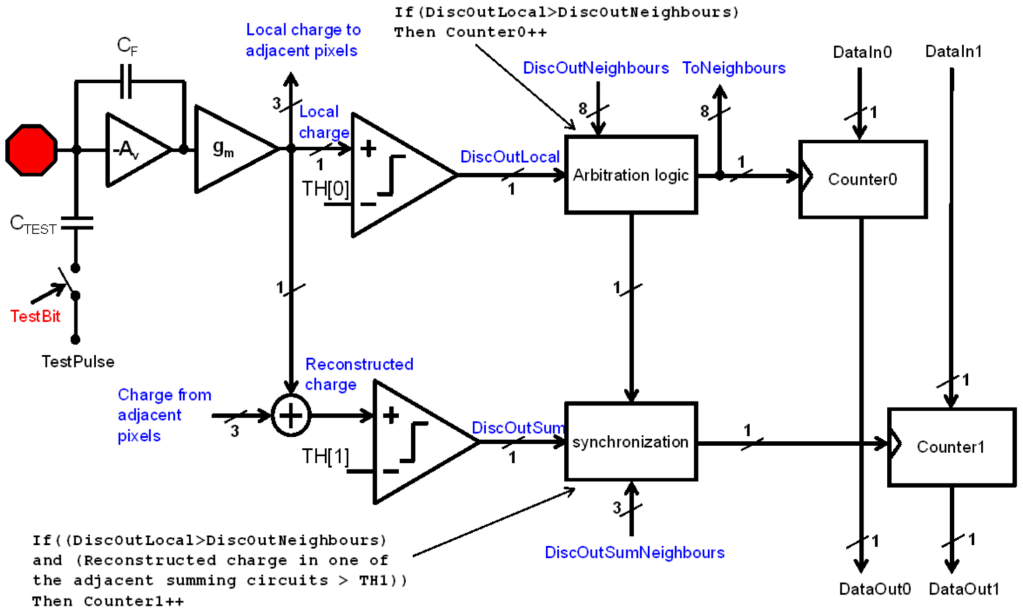


Figure 3. Schematic block diagram of the pixel when configured in Charge Summing Mode.

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Medipix3 counts individual X-rays

Medipix3 is a CMOS readout chip bonded pixel by pixel to a separate semiconductor sensor.

$$\text{X-ray} \longrightarrow \text{charge in sensor} \longrightarrow \text{pulse in readout pixel}$$

If the pulse crosses a threshold, the pixel adds one to a counter. An energy threshold rejects small pulses. Two thresholds can also define an energy range.

A fine-pitch chip has 256×256 pixels on a 55 μm pitch, about 14 mm across.

How much charge does one X-ray produce?

An X-ray deposits energy in the sensor and creates **electron–hole pairs**. The mean energy per pair, W, is about 3.6 eV in silicon and 4.4 eV in CdTe.

$$\bar{N}_{\text{pairs}} = \frac{E_{\text{deposited}}}{W}, \qquad |Q| = e\,\bar{N}_{\text{pairs}} \quad \text{with full charge collection.}$$

Example: If a 50 keV X-ray deposits all its energy in silicon, it creates about $14{,}000$ pairs. The collected signal is about 2.2 fC.

A larger energy deposit gives a larger charge pulse. This is how a threshold on charge can correspond to a threshold on energy.

Charge sharing across pixels

Suppose a 50 keV X-ray deposits all its energy in silicon. The charge cloud spreads across two pixels, which collect charge equivalent to 30 and 20 keV.

At a threshold equivalent to 35 keV, two independent pixels would register zero counts!

Medipix3 can combine signals from nearby pixels and assign one count. This helps recover the event and estimate its energy.

Quantum efficiency (QE)

In a simple counting detector, the **quantum efficiency** η is the fraction of incident photons registered, on average.

$$\bar{N} = \eta \, \bar{N}_{\text{incident}}.$$

If 400 photons reach a pixel on average and $\eta = 0.50$, then the mean detected count is 200.

Photon counts and SNR

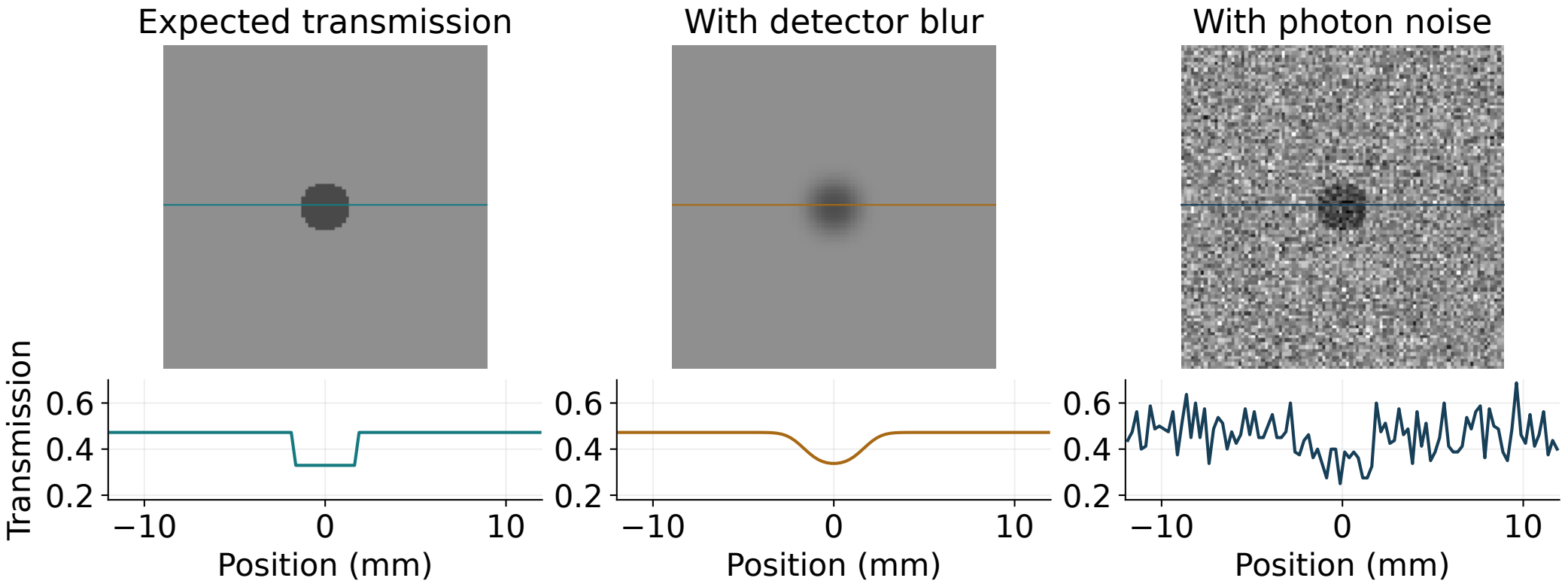
For independent arrivals and detection in this model,

$$\sigma_N = \sqrt{\bar{N}}, \qquad \frac{\sigma_N}{\bar{N}} = \frac{1}{\sqrt{\bar{N}}}.$$

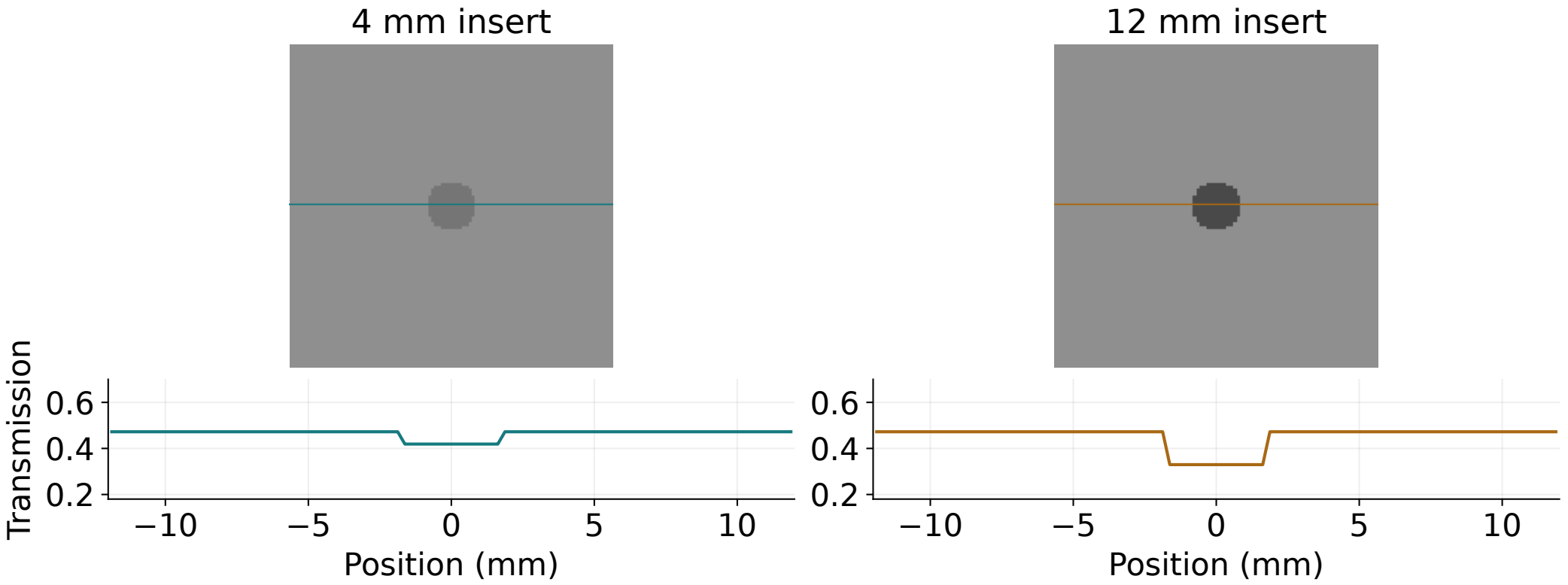
Mean detected count	Standard deviation	SNR
100	10 counts	10 = 20 dB
400	20 counts	20 = 26 dB

Which exposure has the larger fluctuation relative to its mean?

Contrast, blur, and noise change different features

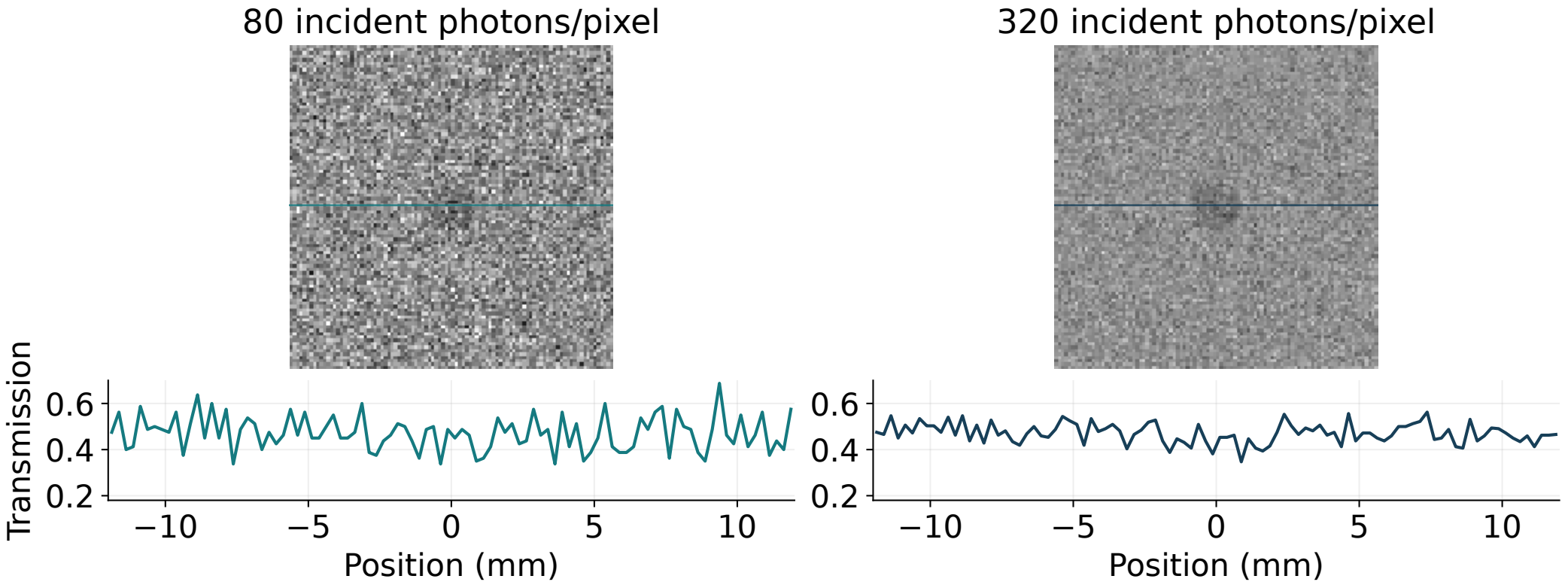


Change the feature thickness



What did we disucss last time that explains the attenuation. Here the sample thickness varies.

Impact of SNR on visibility



The contrast and object are the same. Just more photons on the right.

What detects a dental X-ray today?

Intraoral receptor	How it makes the image
Scintillator + CMOS	A layer such as CsI(Tl) makes light. The CMOS array reads it immediately. One commercial sensor uses $20\ \mu$m pixels.
Storage phosphor plate	A thin plate stores the image. A separate laser scanner reads it after exposure.
Direct-conversion sensor	A semiconductor converts X-rays into charge. Newer commercial examples use a separate absorbing layer, such as a lead-halide photoconductor, on a silicon readout.

Commercial intraoral sensors

Model	Conversion	Pixel pitch	Active area
DEXIS Titanium	CsI scintillator + CMOS	$19.5\ \mu$m	33×26 mm
DentiMax OpenSensorX (size 1)	CsI(Tl) scintillator + CMOS	$20\ \mu$m	30×20 mm
EzSensor XHD (size 1.5)	Direct $CsPbBr_3$ on Si	$14.8\ \mu$m	about 33×24 mm

Discuss: If two sensors have the same pixel pitch, what else would you measure to compare how well they show a small feature?

Detector efficiency reduces exposure

Suppose we need the same mean detected count to see a feature. Compare two ideal counting detectors with efficiencies of 40% and 60%.

$$\bar{N}_{\text{incident}} = \frac{\bar{N}_{\text{detected}}}{\eta}$$

What fraction of the original incident photons does the more efficient detector need?

With the same beam quality and geometry, reducing tube output can reduce patient dose. Keeping the beam limited to the area of interest also helps.

What switches a dental X-ray beam on and off?

Many intraoral units use a **stationary tungsten target**. Short exposures and cooling time between exposures limit heating.

The filament supplies electrons. A high-frequency generator applies the tube voltage for a timed interval. X-rays are produced while accelerated electrons strike the target.

The Planmeca ProX offers 60–70 kV, 2–8 mA, and exposure settings from 0.01 to 2 s.

Specifications and impact

A small feature has weak contrast and appears noisy.

- What would more detected photons change?
- What would a narrower PSF change?

What happens when two structures overlap along the same ray?

Reading and image sources

IAEA, *Diagnostic Radiology Physics*, chapters 4 and 7.

- Gruner, Tate and Eikenberry, CCD area X-ray detectors
- NIST silicon attenuation coefficients
- CERN Medipix3
- ADA dental radiography and radiation safety

The transmission examples are simulations of a uniform background with a small insert.

A commercial camera tiles Medipix3 chips (aside)

The **ADVACAM WidePIX CHROMATIC 25** uses a 5×5 array of Medipix3 chips. It is a photon-counting camera for applications such as industrial X-ray imaging.

Specification	Published value
Image matrix	1280×1280 pixels
Nominal interior pixel pitch	$55\ \mu$m
Sensor options	300 or $500\ \mu$m Si, or $1000\ \mu$m CdTe
Counter depth	12 or 24 bits, depending on mode